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Research report

Blue but not red light stimulation in the dark has antidepressant effect in behavioral despair

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ABSTRACT

The present study investigated potential antidepressant effect of light exposure in the dark phase of a 12:12 L/D cycle on behavioral despair. In Exp.1, male Wistar rats were administered a single, 10 min broadband light pulse (1300 lx) either 3 h (ZT15) or 9 h (ZT21) after dark onset (ZT12) and tested in two consecutive swim tests separated by 24 h. Photic stimulation at ZT21 but not ZT15 significantly reduced immobility in the second swim test relative to the first test compared to controls that were treated similarly except for light pulse administration ($p < 0.05$). In Exp.2, groups were exposed to a single 10-min light pulse (1300 lx) either in the blue or red end of the spectrum at ZT21 or were treated similarly except for photic exposure (controls). Exposure to blue light resulted in significantly reduced duration of immobility in the second swim test relative to that of the first test compared to the red light and control groups ($p < 0.05$). The present findings suggest a critical role in the antidepressant effect of blue light stimulation for the melanopsin-containing ganglion cells in the retina that are sensitive to wavelengths in the blue but not red end of the visible spectrum.

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1. Introduction

Light therapy in the form of bright light exposure either early in the morning or in the evening is effective in treating both seasonal [1,18,19,27] and nonseasonal [16,23] depression. A major target for the antidepressant effect of light administration is the suprachiasmatic nucleus (SCN), the master pacemaker of the brain that regulates a large number of circadian rhythms [45]. The SCN is implicated in affective functions and disorders [10,20,21,39,40,44].

Clinical insight gained by light therapy may benefit from animal models that can provide the basis for elucidating the central mechanisms, including the SCN, that mediate the ameliorative effect of light treatment on depression. There is, however, a paucity of studies that have directly tested the effect of light treatment on depression in animals. A study [24] showed that prolonged exposure of male rats to a long photoperiod (14:10 L/D cycle) has an antidepressant effect as measured by reduced immobility in a forced swim test. Conversely, exposure to a short-day regimen has depressogenic effects in rats [30], hamsters [31] and the sand rat [9].

We have been investigating the effect of light administration in the dark phase of a daily L/D cycle on behavioral despair, an animal model of depression. The model is based on two forced swim tests separated by 24 h [29]. Many antidepressants used to treat depression in humans decrease while depressogenic treatments increase immobility in the second swim test compared to the first test [7,22,29,34]. Earlier, we reported that in female Wistar rats a single 12-h exposure to light in the dark phase of a 12:12 L/D lighting cycle provides immunity to behavioral despair [51]. A realistic animal model of light therapy should be able to replicate in animals the clinical findings related to the time of administration of light exposure. In light therapy, morning light exposure has been shown to be more effective than an equivalent treatment in the evening [15,18,42,50]. Light exposure early in the night results in phase delays, whereas light delivered late at night causes phase advances of the SCN [8,36,46]. Since morning light treatment is found to be superior to evening light treatment, phase advances rather than phase delays by light exposure may be more effective in antidepressant action of light exposure. In view of these findings, we investigated the potential involvement of the SCN in depression by assessing behavioral despair after brief light exposure at different times in the dark phase of an L/D cycle. We showed recently that a 30-min pulse of light delivered late but not early in the dark phase of a 12:12 L/D cycle has antidepressant effect on female Wistar rats as measured by forced swim tests [38]. The present study expands our previous findings by showing that an even shorter light pulse (10 min) has antidepressant effect in male rats when delivered late but not early in the dark phase of a similar lighting schedule.

Furthermore, since our finding parallels clinical reports on light treatment, we aimed to determine whether the effective late pulse shows spectral sensitivity. With the discovery of intrinsically light

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sensitive retinal ganglion cells containing melanopsin [32], new insight has been gained on light input to the SCN. Melanopsin-containing ganglion cells provide the major input to the SCN via the retinohypothalamic tract and have maximum spectral sensitivity in the blue region of the visible spectrum [26]. Studies with rodents and humans show that the most effective spectral wavelengths are between 450 and 550 nm for eliciting phase changes and melatonin suppression [12]. A comparison of the antidepressant effects of blue light at 468 nm and red light at 654 nm in humans demonstrated that light at the blue end of the spectrum was much more effective [11]. Our findings indicate that the differential ameliorative effect of late pulses is likely to be due to their different impact on the SCN. Since melanopsin-containing cells in the retina are more sensitive to the wavelengths in the blue rather than the red end of the visible spectrum and constitute the major input to the SCN, it was hypothesized that a brief pulse of blue light delivered in the dark phase of the 12:12 L/D cycle would have a stronger antidepressant effect on behavioral despair than a pulse of red light.

2. Materials and methods

The study was conducted in two phases in which animals were exposed to a 10-min light pulse (broadband light in Exp.1, blue or red light in Exp.2) late in the dark phase of an L/D cycle followed by forced swim tests.

2.1. Subjects

A total of 48 experimentally naïve male adult Wistar rats bred in the Psychobiology Laboratory were used in the experiments. Weight range of the rats was between 200 and 230 g at the beginning of the experiment. Animals in the experiments were group housed in a vivarium maintained on a standard 12:12 L/D cycle (lights off at 19:00, designated as zeitgeber time 12/ZT12) cycle at approximately 22 ± 2 °C except on the day of experimental manipulation as noted below. Food and water was available ad libitum.

2.2. Apparatus and photic stimuli

Photic stimulation was administrated in a sound and light proof ventilated chamber (60 cm × 60 cm × 70 cm). The daily lighting schedule of the chamber was identical to that of the vivarium (12:12 L/D) except for a brief light pulse administration in the dark phase of the L/D cycle as described below. In the first experiment, light stimulation was provided by a tungsten lamp, while in the second experiment, blue and red pulses of light was provided by filtering the output of six 15 W fluorescent lamps placed approximately 50 cm above the subjects' eye level. Spectral characteristics of the blue and red light pulses are shown in Fig. 1. Intensity of light pulses was kept at approximately 1300 lx in both experiments for all light exposures.

2.3. Forced swim tests (FSTs)

Animals were tested between 14:00 (ZT7) and 16:00 h (ZT16). FSTs consisted of two consecutive days of testing conducted in a Plexiglas cylinder of 45 cm height and 30 cm diameter filled each time with fresh water (25°) to a height of 25 cm. Animals were then allowed to swim for 15 min on the first day of the experiment, followed 24 h later by a second swim test lasting 5 min. Total duration of immobility was recorded in the first 5 min of the first swim test and the entire second test. Immobility was recorded when the animals stayed motionless or floated by keeping the head above the water surface without leaning against the wall of the cylinder.

2.4. Experimental procedure

2.4.1. Experiment 1: effects of broadband light stimulation early and late in the dark phase of the L/D cycle on forced swim tests

Twenty-four adult rats were randomly assigned to one of three groups ($n = 8$ each) for behavioral testing. For light as well as control treatments, subjects were taken from their home cages 10 min before the beginning of the dark period in the L/D cycle and were introduced into the insulated chamber. Four rats in two cages were run each time. The chamber was kept dark in the 12 h dark phase of the L/D cycle (19:00–07:00 h) except for a 10-min, 1300 lx light pulse exposure at either ZT15 (22:00 h; 3 h after lights off) or ZT21 (04:00 h; 9 h after lights off). Control animals were treated similarly except that they were not exposed to a light pulse. For all groups, light was turned on at 07:00 h in the insulated chamber to start the light phase of the L/D cycle. Animals were kept in the insulated chamber until 14:00 h (ZT8) when the first of the forced swimming tests was conducted. Animals were then returned to their home cages in the vivarium where they were kept until the second forced swim test conducted 24 h later.

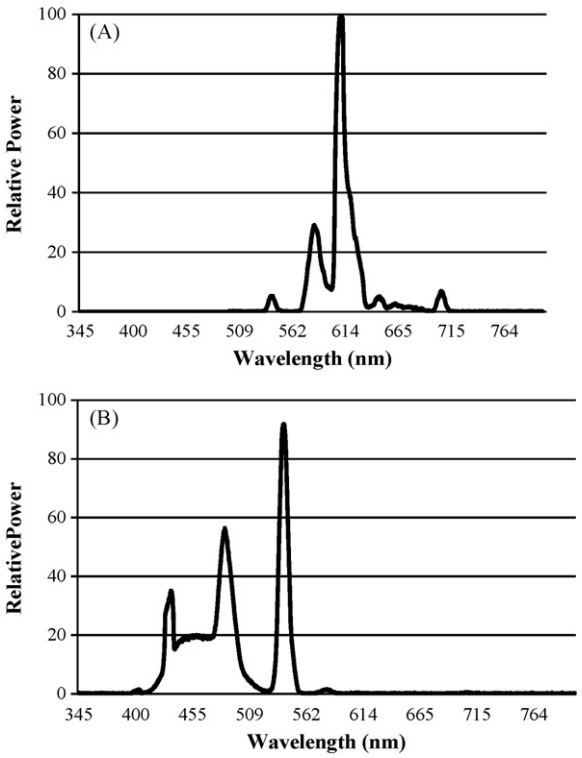


Fig. 1. Spectral power curves for the red (A) and blue (B) light pulses used in Exp.2.

2.4.2. Experiment 2: effects of photic stimulation with different wavelengths on forced swim tests

Twenty-four rats were randomly assigned to two experimental and a control group ($n = 8$ for each). Animals in this experiment were treated similar to those in Exp.1 including administration of FSTs except that they were exposed to a 10 min pulse of either blue or red light 9 h after lights off (at ZT21) or were treated similarly except for photic stimulation (controls).

2.5. Statistics

In behavioral despair, increased immobility in the second swim test compared to the first is a sign of behavioral despair while treatment-induced reduction of immobility in the second test is an indication of antidepressant action [29]. Accordingly, the data in both experiments were analyzed in terms of the percentage of increase in immobility in the second swim test compared to the first (means ± S.E.M). The first 5 min of the first test and the entire second test were evaluated for these comparisons. An analysis of variance (ANOVA) was used to compare the effect of treatment followed by Newman–Keuls tests as post hoc evaluation of significant differences of $p < 0.05$ or better.

3. Results

3.1. Experiment 1

Fig. 2A depicts percentages of change in immobility in the second swim test compared to the first test for the groups receiving light treatment at ZT15 or ZT21 and the control group. ANOVA indicated a significant group effect [$F(2, 21) = 3.99$, $p < 0.05$] which was due to the significant difference between the group receiving light treatment at ZT21 compared to controls ($p < 0.05$). The ZT15 group did not differ from control animals.

3.2. Experiment 2

Fig. 2B shows the percentage of change in immobility from the first swim test to the second for the groups receiving blue or red light stimulation at ZT21 as well as the control group. ANOVA indicated a significant group effect in percentages [$F(2, 21) = 4.47$, $p < 0.05$]. Post hoc tests showed that blue light group was sig-

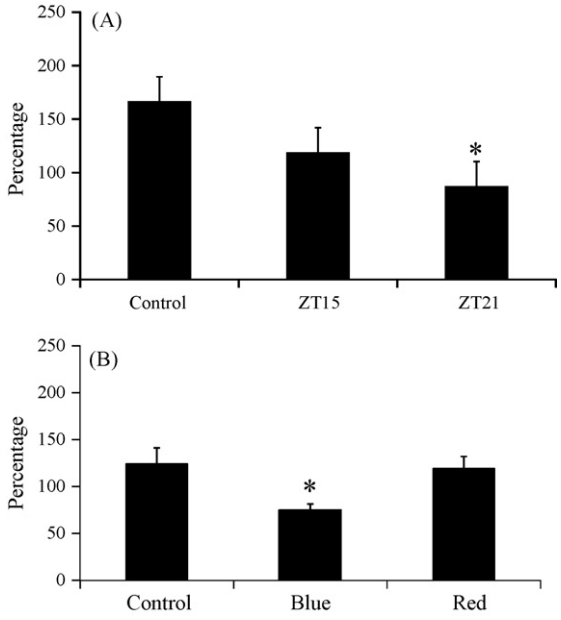


Fig. 2. Percentages of change in duration of immobility in the second forced swim test compared to the first 5 min of the first swim test (mean ± S.E.M.). Groups in A received a 10 min pulse of light (1300 lx) in the dark phase of a 12:12 L/D lighting cycle either at ZT15 (3 h after lights off at ZT12) or at ZT21 (9 h after lights off). In B, groups received either a blue or a red pulse of light (10 min, 1300 lx) at ZT21. In both graphs, control animals underwent similar treatments as the experimental groups except for exposure to a light pulse. *Significant difference ($p < 0.05$) between the ZT21 group and controls in A and between the blue pulse group and the red pulse and control groups in B.

nificantly different from both the red light ($p < 0.05$) and control ($p < 0.05$) groups, while the control and red light groups were not significantly different from each other.

4. Discussion

The present study showed that a 10 min pulse of broadband light (1300 lx) has antidepressant effect as measured by immobility in forced swim tests when presented late (at ZT21) but not early (at ZT15) in the dark phase of a 12:12 L/D light cycle. Differential ameliorative effect of broad band light stimulation late but not early in the dark phase of the L/D cycle may have resulted from the fact that photic stimulation at different times in the dark phase of the L/D cycle can affect the suprachiasmatic nuclei (SCN) differentially. It is well established that light administration early in the dark phase delays and late in the dark phase advances the regulation of behavioral and biological rhythms by the SCN in both humans and rodents [13,25]. Our findings therefore parallel clinical reports that light therapy early in the morning is more efficient than that in the evening [15,18,42].

A major finding of the present study is that blue but not red light stimulation was found to have antidepressant effect in behavioral despair applied late in the dark phase of the 12:12 L/D cycle. To the best of our knowledge, this is the first demonstration of the differential effect of blue light in an animal model of depression. The present findings are in agreement with the report [11] that the antidepressant effect of blue light of 468 nm wavelength was superior to that of red light at 654 nm in humans. Our findings point to the relevance of melanopsin-containing retinal ganglion cells to the ameliorative effect of light therapy. Melanopsin-containing ganglion cells are considerably more sensitive to short wavelengths of light compared to light in the red end of the visible spectrum and have an essential role in the regulation and phase shifting of circadian rhythms [26]. Importantly, a recent study with depressed patients reported that a missense variant of the melanopsin (OPN4) gene may increase the risk for Seasonal Affective Disorder in some individuals [35].

The central mechanism by which light stimulation affects depressive symptoms is not fully understood. Light is the major input to the SCN that regulates a large number of circadian rhythms including that of melatonin [25,43,45]. Therefore, a phase shift by means of either broad band or blue light stimulation can have the dual impact of altering the circadian rhythmicity of melatonin secretion by the pineal gland and that of many other biological functions that are regulated by the SCN. A brief disruption of melatonin secretion or its rhythmicity as in the present study, however, is not likely to explain the present findings in that administration of melatonin or its agonists has been shown to have an antidepressant effect in rodents in forced swim tests [2,3,28,33].

Instead of altering melatonin secretion or rhythmicity, the antidepressant effect of light pulses in the present study may be due to a cascade of events triggered by light-induced SCN activation in the dark phase of the L/D cycle. There is growing neurochemical as well as neuroanatomical evidence that the SCN is part of a neural system that is closely related to affective behavior and disorders [21,41]. Neurochemically, involvement of the SCN in depression may be related to its role in the regulation of the hypothalamo–pituitary–adrenal (HPA) axis response to stress [5,6,14,37]. Neuroanatomical studies on the other hand indicate neural connectivity between the SCN and several limbic structures involved in affective behavior and psychopathology [4,17,47–49].

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